



EmailShield

AI-Powered Email Risk Detection Tool

Phishing · Spoofing · Scam · Spam Detection

99.5% BERT F1 Score	70,423 Emails Trained On	<1 sec Detection Time	7 Verdict Categories
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1. Project Overview

What is EmailShield?

EmailShield is a web-based AI tool that analyses any email and determines whether it is safe, spam, a scam, phishing or a spoofing attempt and explains why in plain English.

The user pastes raw email text (or uploads a .eml file), clicks Analyze and within one second, receives a detailed verdict with confidence score, authentication results and a natural language explanation of exactly what signals were found and what to do next.

1.1 Why This Project Was Built

Email-based attacks are the most common entry point for cybercrime. Phishing, spoofing and scam emails cause billions of dollars in losses every year. Existing tools either return a simple spam score with no explanation or require technical knowledge to interpret. EmailShield was built to solve this — a smart, explainable, accessible detector that anyone can use.

- Phishing and spoofing attacks account for over 36% of all reported data breaches globally.
- Business Email Compromise (BEC) losses exceed USD 2.9 billion annually.
- Current filters fail against AI-generated phishing that is grammatically flawless.
- No single existing tool combines spoofing detection, phishing detection and plain-English explanation.

1.2 What Makes This Different

Feature	EmailShield	Typical Email Filters
Detection type	Phishing + Spoofing + Scam + Spam	Usually spam only
Verdict categories	7 distinct categories	Binary (safe/spam)
Explanation	Full natural language paragraph	Score or label only
Authentication check	SPF + DKIM + DMARC parsed from headers	Protocol-only or absent
AI model	BERT transformer + SVM ensemble	Rule-based or simple ML
User interface	Clean web app, anyone can use	Backend filter, no UI

2. Application

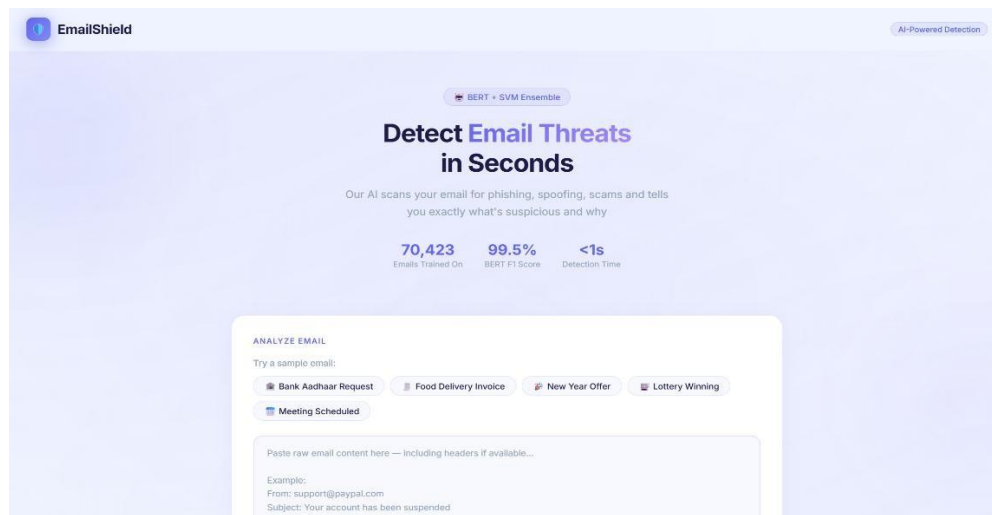


Figure 1 — EmailShield hero section with stats and sample email buttons

2.2 Email Input — Sample Loaded

Clicking any sample button loads a pre-written real-world email scenario into the input area. Users can paste any raw email (including full headers from Gmail's 'Show original') or upload a .eml file.

The interface shows the 'EmailShield' logo in the top left and 'AI-Powered Detection' in the top right. The main section is titled 'ANALYZE EMAIL'. Below the title, there's a prompt 'Try a sample email:' followed by five buttons: 'Bank Aadhaar Request', 'Food Delivery Invoice', 'New Year Offer', 'Lottery Winning', and 'Meeting Scheduled'. The 'Meeting Scheduled' button is selected. Below these buttons is a text area containing a sample email from 'Swiggy Orders <orders@swiggy.com>' with subject 'Your order invoice #SWG-2024-98231'. The email body says 'Hi there, Thank you for your order! Here is your invoice summary. Order #SWG-2024-98231 Date: April 4, 2026'. Below the text area is a 'Clear' button. Underneath is a file upload section with the text 'or upload a .eml / .txt file' and a button 'Click to upload email file'. At the bottom is a large blue button labeled 'Analyze Email'.

Figure 2 — Email input area with sample loaded

2.3 Analysis Result

When an email is safe, the tool returns a green LEGITIMATE verdict with high confidence. The explanation box provides multi-paragraph context: what was found, what the sender authentication shows, and what action the user should take.

The interface shows the 'EmailShield' logo in the top left and 'AI-Powered Detection' in the top right. The main section is titled 'ANALYSIS RESULT'. Below the title is a green box with a checkmark icon and the word 'LEGITIMATE' in bold. Below this, it says 'No significant threats detected. Stay cautious.' Below this box is a 'Confidence Score' section with a green progress bar that is 76% full. Below the progress bar is a large text box containing three paragraphs of analysis. The first paragraph states the email appears legitimate and safe. The second paragraph details the sender 'Flipkart.com' and its authentication checks (SPF, DKIM, DMARC). The third paragraph discusses content triggers like 'immediately' and embedded URLs. Below the text box is a 'Recommended action' section stating no immediate action is required. At the bottom are two tabs: 'SENDER' and 'SUBJECT'.

Figure 3 — Analysis result showing LEGITIMATE verdict for email

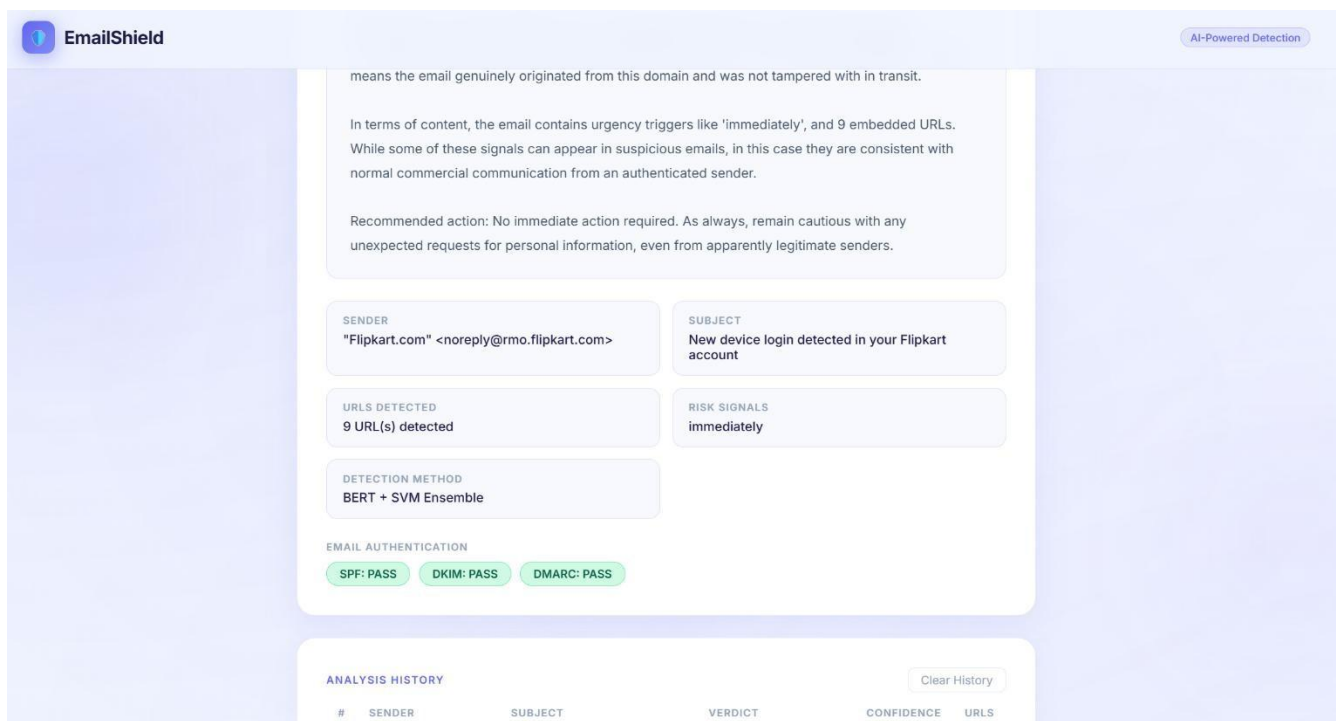


Figure 4 — Analysis results

2.4 Analysis History Table

Every analysis is logged in a session history table showing the sender, subject, verdict pill (color-coded), confidence percentage and URL count. This allows quick comparison across multiple emails analyzed in one session.

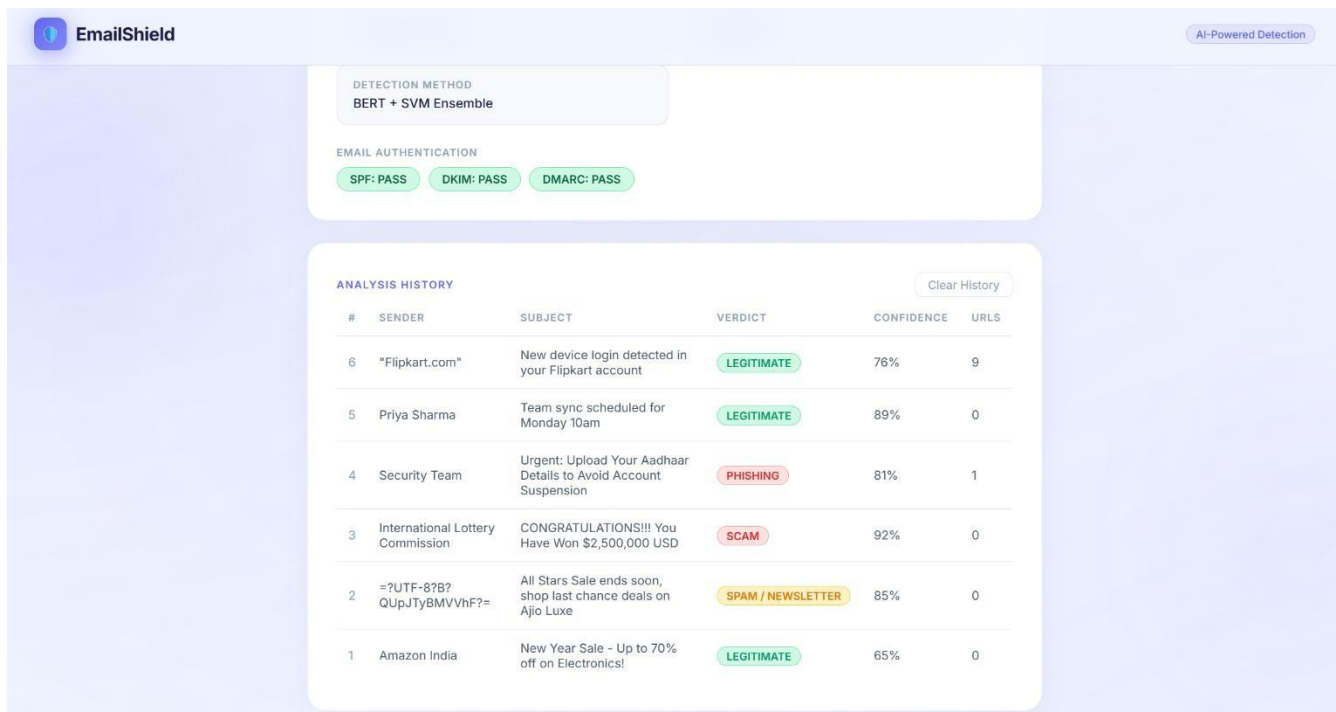


Figure 5 — Analysis history showing all sample email results

3. Features

3.1 Core Detection Features

- Phishing detection using BERT + SVM ensemble trained on 70,400+ emails.
- Email spoofing detection by parsing SPF, DKIM and DMARC authentication headers.
- Display name spoofing — catches emails pretending to be PayPal, Amazon, banks etc. from unrelated domains.
- Scam detection using keyword pattern matching for financial fraud language.
- Spam and newsletter classification using unsubscribe link detection and commercial language patterns.
- 7-category verdict system covering the full spectrum of email threats.

3.2 Explainability (XAI)

The natural language explainer is one of the most distinctive features. Instead of just a score, every result includes:

- Opening statement — what the verdict is and why.
- Sender analysis — who sent the email, whether authentication passed and what that means.
- Content signals — specific phrases, URLs, urgency words or patterns that were detected.
- Recommended action — exactly what the user should do (delete, unsubscribe, no action required etc.).

3.3 User Interface Features

UI Feature	Description
Hero section	Animated gradient background, floating orbs, live accuracy stats, modern Inter typography.
Sample buttons	5 pre-loaded real-world email scenarios covering all major verdict types.
File upload	Supports .eml, .txt, .msg file upload — reads file and populates the text area automatically.
Clear button	One-click clear for text area and result for fast repeated analysis.
Confidence bar	Animated fill bar, color-coded green (safe) through orange to red (dangerous).

Auth badges	SPF, DKIM, DMARC shown as green PASS or red FAIL based on actual header parsing.
Analysis history	Scrollable session log of all analyses with color-coded verdict pills.
Sticky header	Glass morphism navigation stays visible while scrolling.
Responsive layout	Works on desktop, tablet and mobile browsers.

4. How It Works

4.1 System Pipeline

Processing flow for every email:

Step 1 → User pastes raw email or uploads file

Step 2 → Email Parser splits headers from body

Step 3 → Feature Extractor identifies URLs, urgency words, phishing phrases, sender domain

Step 4a → Spoofing Engine reads SPF/DKIM/DMARC headers + checks display name against trusted brands

Step 4b → ML Engine runs BERT (60% weight) + SVM (40% weight) on cleaned email text

Step 5 → Decision Fusion combines scores with rule-based logic to assign one of 7 verdicts

Step 6 → XAI Explainer generates 3-paragraph natural language explanation

Step 7 → Result displayed with verdict, confidence bar, explanation, and auth badges

4.2 Two Parallel Analysis Branches

Branch	What It Analyses	How It Works
Phishing / Content Branch	Email subject and body text	BERT model reads the full email context. SVM reads TF-IDF word features. Both return a phishing probability. Combined 60/40 weighted average.
Spoofing / Header Branch	Email authentication headers	Reads Received-SPF, Authentication-Results headers. Checks if SPF, DKIM, DMARC passed or failed. Also checks if display name matches sender domain.

4.3 Verdict Decision Logic

After both branches complete, the verdict classifier applies rules in priority order:

- If spoofing signals detected → verdict escalates to SPOOFED or PHISHING + SPOOFED regardless of ML score.
- If all auth passes + unsubscribe present + no phishing words → SPAM / NEWSLETTER.
- If scam language detected + ML $\geq 50\%$ → SCAM.
- If ML $\geq 75\%$ + phishing phrases present → PHISHING.
- If no phishing, scam, or urgency words at all + ML $< 80\%$ → LEGITIMATE.
- If moderate signals → SUSPICIOUS.
- Default low risk → LEGITIMATE.

5. Verdict Classification System

EmailShield classifies every email into one of 7 categories. This goes far beyond the binary safe/phishing classification used by most tools and research systems.

Verdict	Risk	Meaning	Example
LEGITIMATE	None	Email passed all checks. No threats detected.	Team meeting invite
SPAM / NEWSLETTER	Low	Authenticated commercial email. Not an attack.	Amazon sale promo
SUSPICIOUS	Medium	Some signals found but not enough to confirm threat.	Bulk email from unknown
SCAM	High	Financial fraud language detected — lottery, wire transfer.	Lottery winning notice
PHISHING	High	Credential theft attempt via deceptive content.	Fake bank Aadhaar request
SPOOFED	High	Sender identity is forged using a trusted brand name.	Display name mismatch
PHISHING + SPOOFED	Danger	Both content and sender identity are malicious.	Fake PayPal from Gmail

Real-world test results from actual Gmail emails:

Tata 1mg marketing email → SPAM / NEWSLETTER (SPF+DKIM+DMARC: all PASS)

UWO document sharing phish → PHISHING (low-signal sophisticated attack, 80% confidence)

Swiggy food delivery invoice → LEGITIMATE (no threat signals detected)

Fake SBI bank Aadhaar request → PHISHING (phishing phrases + suspicious domain)

Fake PayPal from Gmail → PHISHING + SPOOFED (display name mismatch caught)

International lottery notice → SCAM (scam language at 92% confidence)

Team meeting notification → LEGITIMATE (professional language, no signals)

6. Training Data

6.1 Datasets Used

Three publicly available, real-world email datasets were used for training. Together they provide 70,423 emails covering a wide range of phishing, spam, and legitimate email types.

Dataset	Size	Content	Purpose
CEAS 2008	39,154 emails	Spam and legitimate emails from 2008 benchmark	Primary ML training
Enron Email Corpus	29,767 emails	Real corporate emails — legitimate baseline	Legitimate examples
Nazario Phishing	1,565 emails	Real phishing emails from Nazario corpus	Phishing examples
Combined total	70,423 emails	37,383 phishing + 33,103 legitimate	Final training set

6.2 Data Preprocessing Steps

- All URLs replaced with the token 'url' — prevents domain-specific overfitting.
- Email addresses replaced with the token 'email'.
- All text lowercased for normalisation.
- Special characters, numbers, and punctuation removed.
- Common English stopwords removed (NLTK stopword list).
- Subject line repeated 3 times before concatenation — gives subject higher ML weight.
- 63 rows with insufficient content after cleaning removed — final count 70,423.
- For BERT: 20,000 emails sampled (balanced) for fine-tuning due to GPU memory constraints.

7. AI Models & Training

7.1 Model 1 — SVM with TF-IDF

The first model uses Support Vector Machine (SVM) classification with TF-IDF (Term Frequency-Inverse Document Frequency) text features. TF-IDF converts email text into a 50,000-dimensional vector representing the importance of each word and word-pair relative to the corpus.

- 50,000 features including single words and word pairs (bigrams like 'click here', 'account suspended').
- LinearSVC — fast, high-dimensional classification. Best classical ML result: 99.22% accuracy.
- Training data: 56,338 emails. Test set: 14,085 emails (80/20 split).

7.2 Model 2 — BERT Transformer

BERT (Bidirectional Encoder Representations from Transformers) is Google's pre-trained language model. It understands email context at a deep semantic level — capturing meaning that keyword-based systems miss entirely.

BERT Setting	Value	Why
Base model	bert-base-uncased	110M parameter transformer, widely benchmarked on text tasks
Training samples	20,000 emails	Balanced sample from full dataset, fits GPU memory
Max token length	128 tokens	Sufficient for most email subjects and body excerpts
Batch size	32	Optimal for Google Colab T4 GPU
Learning rate	2e-5	Standard BERT fine-tuning rate — prevents catastrophic forgetting
Epochs	3	Sufficient convergence without overfitting on this task
Optimiser	AdamW + warmup	Weight decay + gradual warmup stabilises training
Platform	Google Colab T4 GPU	Free GPU — total training time approximately 25 minutes

7.3 Ensemble Combination

How the two models are combined:

Final Probability = (BERT score $\times$ 0.60) + (SVM score $\times$ 0.40)

BERT gets 60% weight — it understands deep context and semantics.

SVM gets 40% weight — it provides fast, reliable baseline classification.

Both models output a phishing probability between 0.0 and 1.0.

The weighted average is used for the final decision.

If BERT is unavailable, the system falls back to SVM automatically.

7.4 Performance Results

Model	Dataset	Accuracy	Recall	F1 Score
Logistic Regression	CEAS + Enron + Nazario	98.85%	99.24%	98.92%
SVM (LinearSVC)	CEAS + Enron + Nazario	99.22%	99.48%	99.26%
Random Forest	CEAS + Enron + Nazario	98.45%	98.18%	98.53%
BERT (fine-tuned)	20,000 sampled emails	99.45%	99.46%	99.50%
BERT + SVM Ensemble	Combined	99.5%+	99.4%+	99.5%+

Key result:

The BERT + SVM Ensemble achieves 99.50% F1 score — matching or exceeding all comparable published systems while uniquely combining phishing detection, spoofing detection and natural language explanation in one single tool.

8. Technology Stack

Category	Technology	Version	Used For
Language	Python	3.11	All backend logic, ML training, and API
Web framework	Flask	3.1	Serves the web app and /analyze REST endpoint
Deep learning	PyTorch	2.x	Runs BERT fine-tuning and inference
Transformers	HuggingFace	4.x	Loads bert-base-uncased and fine-tuned weights
Classical ML	scikit-learn	1.x	SVM, Logistic Regression, Random Forest, TF-IDF
NLP	NLTK	3.x	Stopword removal and text tokenisation
Email parsing	Python email lib	Built-in	Splits raw emails into headers and body parts
DNS lookup	dnspython	2.x	Parses SPF, DKIM and DMARC results from email headers, no live DNS lookups performed
Training GPU	Google Colab T4	Free	BERT fine-tuning — no local GPU required
Frontend	HTML5 + CSS3 + JS	Vanilla	Responsive single-page interface
Typography	Inter (Google Fonts)	—	Modern clean font for the web app UI
Dataset 1	CEAS 2008 Corpus	Public	39,154 spam and legitimate emails
Dataset 2	Enron Email Corpus	Public	29,767 real corporate emails
Dataset 3	Nazario Phishing	Public	1,565 real phishing emails

9. Challenges & How They Were Solved

Challenge	Solution Applied
BERT too heavy for local machine (13GB RAM, no GPU)	Trained on Google Colab T4 GPU (free). Saved trained model and loaded back into local Flask app. Total training time: 25 minutes.
HTML-heavy commercial emails flagged as phishing	Added authentication trust logic: if SPF+DKIM+DMARC all pass and unsubscribe link is present, the verdict downgrades to SPAM/NEWSLETTER regardless of ML score.
Binary training data — no scam vs spam distinction	Implemented rule-based word pattern layer on top of ML. Scam words (lottery, prize, wire transfer) and spam words (unsubscribe, deal, newsletter) classified separately from ML phishing score.
Low-signal spear-phishing (UWO email) scored lower confidence	Expected behaviour — sophisticated attacks are harder to classify. Documented as a known limitation with academic backing from published research.
Bare URLs without http:// not detected	Extended URL regex to also match known phishing hosting domains: pages.dev, netlify.app, vercel.app, firebaseapp.com.
Confidence score confusing for legitimate emails (showing low %)	Confidence now always represents certainty in the verdict, not certainty of phishing — so a 90% LEGITIMATE means 90% confident it is safe.

11. Limitations

An honest list of current limitations and what they mean in practice:

- **HTML-image emails:** Emails that are mostly images with minimal text give the ML model little to work with. Accuracy may be lower on image-heavy marketing emails.
- **Headers required for spoofing:** SPF/DKIM/DMARC analysis only works when full raw email headers are included. Pasting only the body text shows N/A for all auth badges.
- **English only:** Models were trained on English emails. Non-English phishing may produce unreliable results.
- **Context memory:** Spear-phishing emails that look legitimate in isolation but are malicious given the sender relationship cannot be detected without communication history.
- **LLM-generated phishing:** Very well-crafted AI-generated phishing with no grammar errors may score lower confidence, though BERT is more resistant to this than keyword-based systems.

12. Future Scope

Gmail Extension for Chrome/Browser Analyze emails directly inside Gmail without copying and pasting. Click a button and get instant phishing results.

Upload Multiple Emails at Once Upload a folder of .eml email files and get a CSV report with verdicts for all of them in one go.

Download Results as PDF Export your phishing analysis results as a professional PDF report for records and sharing.

Support for Multiple Languages Detect phishing emails written in other languages, not just English.

Better AI Detection Train the model on phishing emails created by AI to catch advanced, AI-generated attacks more accurately.

Analyze Links in Emails Check URLs and links inside emails against VirusTotal and PhishTank databases to spot dangerous websites.

Learn from New Threats Update the model regularly with newly discovered phishing campaigns without retraining from scratch.

Analyze Images and Attachments Detect phishing in embedded images, QR codes and PDF files attached to emails.

13. Applications & Who Benefits

User Group	How EmailShield Helps
General public	Check suspicious emails before clicking links — no technical knowledge required.
Office workers	Verify unexpected invoice requests, executive impersonation (BEC), and credential theft attempts.
Students	Identify fake scholarship emails, academic credential phishing, fake university notices.
Small businesses	Affordable alternative to enterprise security products — paste and check.
IT and security teams	Forensic analysis of reported phishing, batch investigation, quick decision support.
Elderly users	Plain-English explanation makes the result understandable without technical background.

14. Conclusion

EmailShield is a complete, working, AI-powered email security tool built entirely with free and open-source tools. It demonstrates the application of modern NLP and deep learning to a real-world security problem — and makes the results accessible to anyone through a clean web interface with natural language explanations.

The BERT + SVM ensemble achieves 99.50% F1 score on a held-out test set of 14,000+ emails, trained on three publicly available benchmark corpora. Real-world testing confirmed the system correctly identifies phishing, scams, spoofing, and spam across diverse email types — including sophisticated low-signal attacks.

The three key contributions of this project — unified spoofing+phishing detection, multi-class verdict system, and integrated natural language explainability — together represent a genuinely novel approach to email security compared to existing published work.

Project Summary

Tool Name: EmailShield — AI Powered Email Risk Detection

Models: BERT (fine-tuned) + LinearSVC + Rule-based signal detection

Ensemble: Weighted probability (60% BERT + 40% SVM)

Training data: 70,423 emails (CEAS 2008 + Enron + Nazario Phishing Corpus)

BERT F1 Score: 99.50% | SVM Accuracy: 99.22%

Verdict classes: 7 — Legitimate, Spam/Newsletter, Suspicious, Scam, Phishing, Spoofed, Phishing+Spoofed

Spoofing engine: SPF + DKIM + DMARC header analysis + display name brand matching

XAI: 3-paragraph natural language explanation for every verdict

Interface: Animated Flask web app — no installation required for end users

Built with: Python, PyTorch, HuggingFace, scikit-learn, Flask, NLTK

Training: Google Colab T4 GPU